Week-7-Exercise-Solutions.R

gcrone14

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```
##### Week 7 Exercise Solutions #####
# Load libraries
library(ez)
```

```
## Registered S3 method overwritten by 'lme4':
##   method           from
##   na.action.merMod car
```

```
library(psych)
library(here)
```

```
## here() starts at /Users/gcrone14/Desktop/R Projects/PSYC3031-F26
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x ggplot2::%+%()   masks psych::%+%()
## x ggplot2::alpha() masks psych::alpha()
## x dplyr::filter()  masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

1. Install the cereal 3.csv data into a new R script. You may call this script whatever you wish (e.g., Week7_practice.R). Then, save the cereal data as cereal_dat.

```
cereal_dat <- read_csv(here::here("Data/cereal 3.csv"))
```

```
## Rows: 77 Columns: 16
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (3): name, mfr, type
## dbl (13): calories, protein, fat, sodium, fiber, carbo, sugars, potass, vita...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

2. Perform the following data cleaning steps and save the result into a new data frame called cereal_final. The steps are as follows:

2a. Select the name, shelf and rating variables from cereal_dat.

2b. Change the structure of the shelf variable to a factor and relabel the levels so that: * 1 = Bottom Shelf * 2 = Middle Shelf * 3 = Top Shelf

2c. Change the structure of the name variable to a factor.

We do all three stpes in one go:

```
cereal_final <- cereal_dat |>
  select(name, shelf, rating) |>
  mutate(shelf = factor(shelf,
                        1:3,
                        c("Bottom Shelf", "Middle Shelf",
                          "Top Shelf")
                        ),
         name = factor(name)
         )
```

We verify it works:

```
cereal_final |> glimpse()
```

```
## Rows: 77
## Columns: 3
## $ name   <fct> 100% Bran, 100% Natural Bran, All-Bran, All-Bran with Extra Fib~
## $ shelf  <fct> Top Shelf, Top Shelf, Top Shelf, Top Shelf, Top Shelf, Bottom S~
## $ rating <dbl> 68.40297, 33.98368, 59.42551, 93.70491, 34.38484, 29.50954, 33.~
```

3. Go through the different steps of a One Way ANOVA to determine whether there is a difference in the average rating of cereals placed on the bottom, middle, and top shelf. Write a brief APA summary of the results.

Prior to doing the ANOVA, we should explore the data!

```
# Check frequencies of shelf
summary(cereal_final$shelf)
```

```
## Bottom Shelf Middle Shelf    Top Shelf
##           20           21           36
```

```
# Check rating descriptives
cereal_final |>
  select(rating) |>
  describe()
```

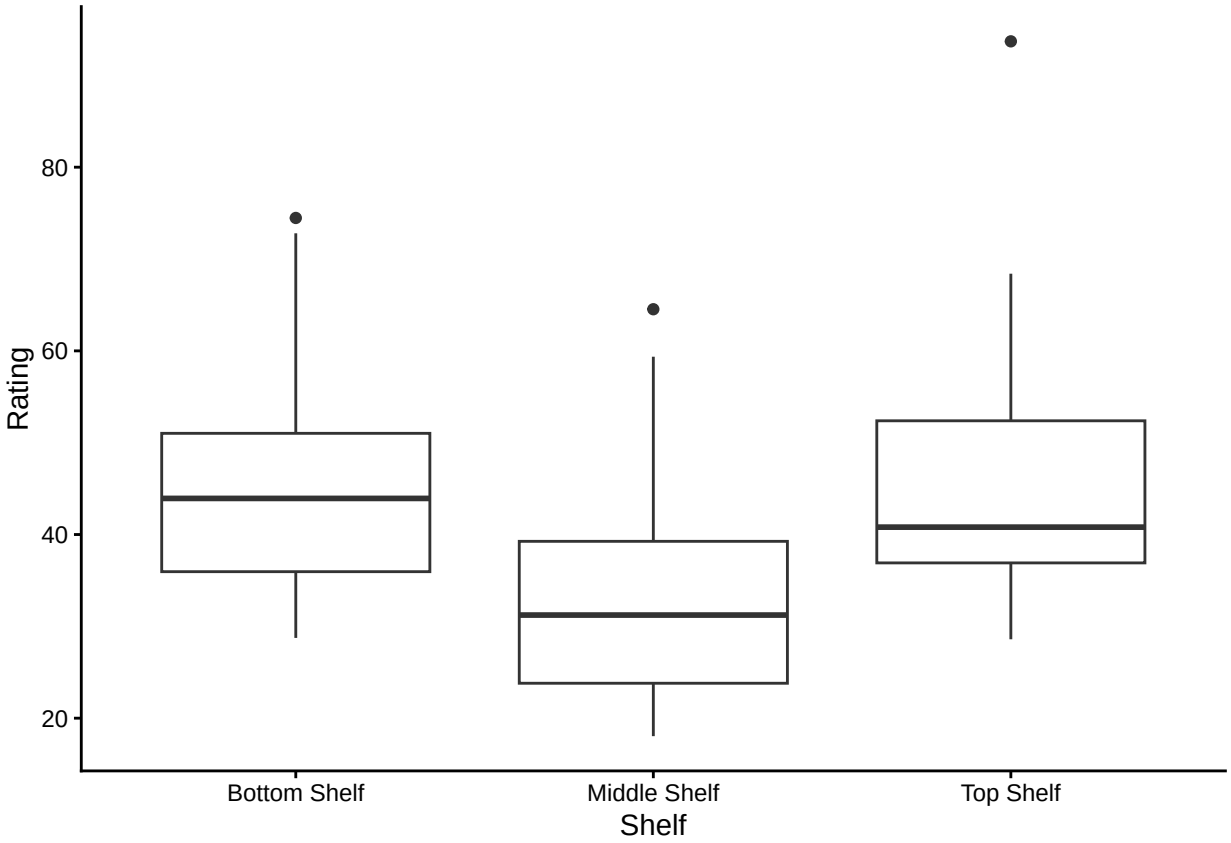
```
##        vars  n  mean    sd median trimmed  mad   min  max range skew kurtosis
## rating    1 77 42.67 14.05   40.4   41.66 13.6 18.04 93.7 75.66 0.88     1.05
##         se
## rating 1.6
```

```r
# Check ratings by shelf
cereal_final |>
  select(rating) |>
  describeBy(cereal_final$shelf)
```

```
##
##  Descriptive statistics by group
## group: Bottom Shelf
##        vars  n  mean    sd median trimmed   mad   min   max range skew kurtosis
## rating    1 20 46.15 13.56  43.93   44.84 11.42 28.74 74.47 45.73 0.68    -0.51
##          se
## rating 3.03
## ------------------------------------------------------------
## group: Middle Shelf
##        vars  n  mean    sd median trimmed  mad   min   max range skew kurtosis
## rating    1 21 34.97 13.95  31.23   33.69 11.9 18.04 64.53 46.49 0.79    -0.71
##          se
## rating 3.04
## ------------------------------------------------------------
## group: Top Shelf
##        vars  n  mean    sd median trimmed  mad   min  max range skew kurtosis
## rating    1 36 45.22 12.97   40.8   43.78 8.85 28.59 93.7 65.11 1.58     3.21
##          se
## rating 2.16
```

Now let's make plots to visualize this mean difference!

```r
cereal_final |>
  ggplot(aes(x = shelf, y = rating)) +
  geom_boxplot() +
  theme_classic() +
  labs(x = "Shelf", y = "Rating")
```

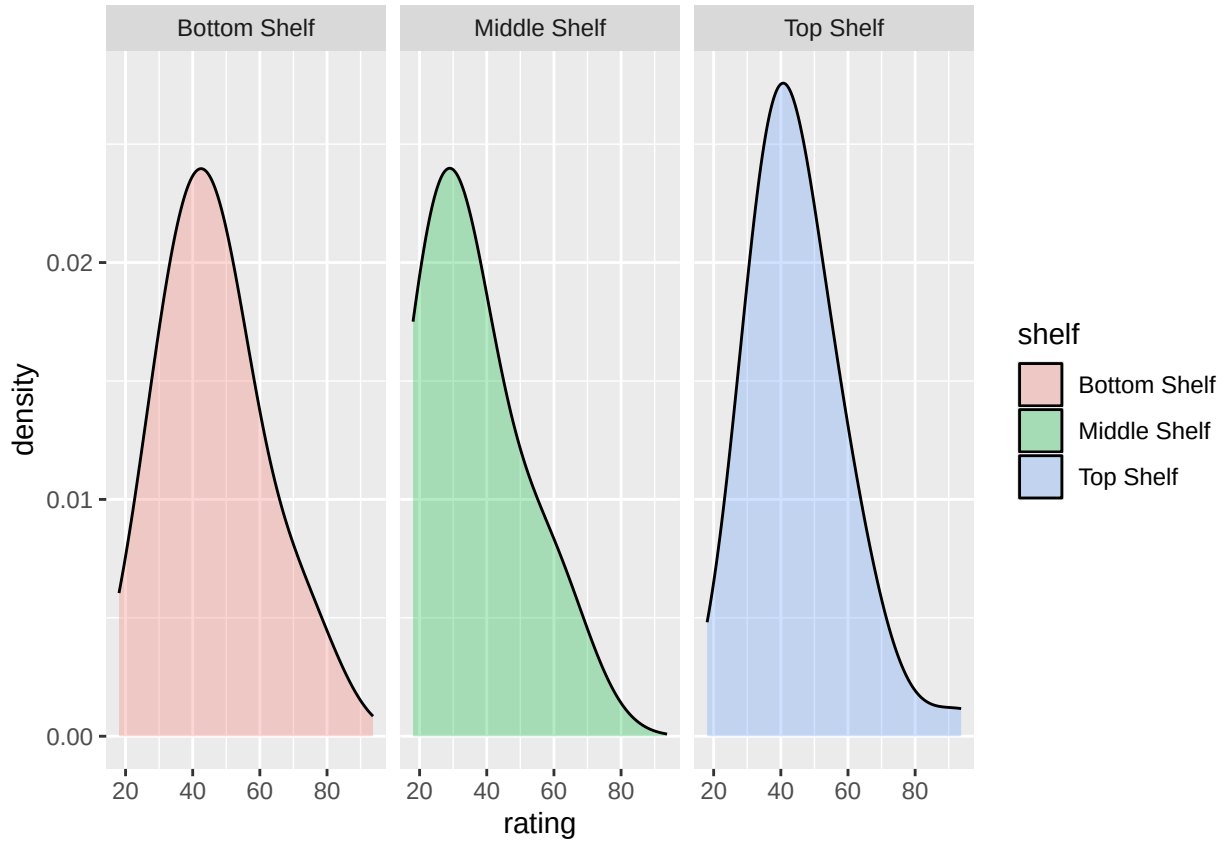


It seems that bottom and top shelf cereals are far more favorably rated than middle shelf ones.

Now, let's check the assumptions

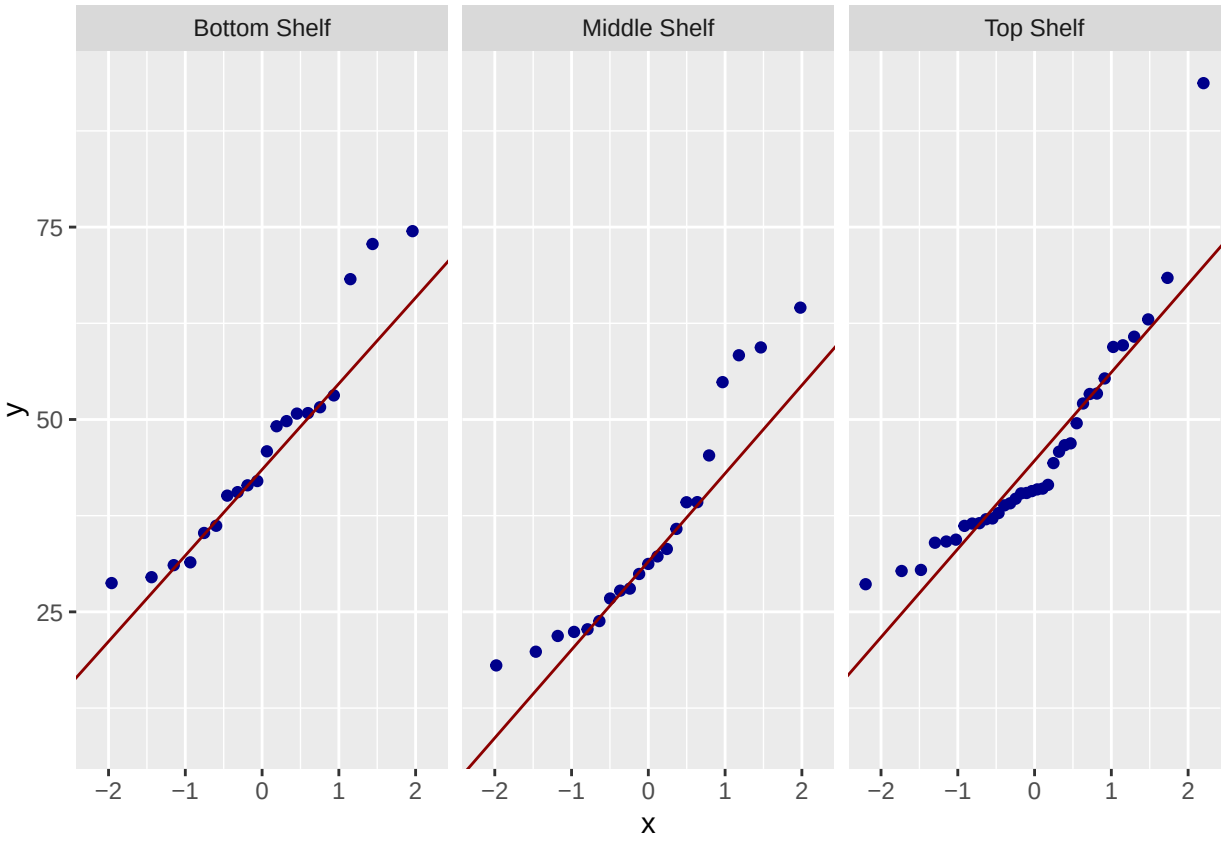
1) Normality

```
cereal_final |>
  ggplot(aes(x = rating, fill = shelf)) +
  geom_density(alpha = 0.3, adjust = 2) +
  facet_wrap(~shelf)
```



Normality doesn't really seem to be met here, but it might be the plot choise that makes it seem this way (deviation doesn't appear too bad). We can also check a Q-Q plot (more on this for Week 8)

```
cereal_final |>
  ggplot(aes(sample = rating)) +
  geom_qq(colour = "dark blue") +
  geom_qq_line(color = "dark red") +
  facet_wrap(~shelf)
```



Good alignment between observed points and the Q-Q line, so seems tentatively met.

2) Homogeneity of variances

```
res <- ezANOVA(data = cereal_final,
               wid = name,
               between = shelf,
               dv = rating,
               return_aov = TRUE)
```

```
## Warning: Data is unbalanced (unequal N per group). Make sure you specified a
## well-considered value for the type argument to ezANOVA().
```

```
## Coefficient covariances computed by hccm()
```

```
res$`Levene's Test for Homogeneity of Variance`
```

```
##   DFn DFd      SSn      SSd         F         p p<.05
## 1   2  74 45.12577 6928.918 0.2409689 0.7864803
```

```
# OR
res[2]
```

```
## $`Levene's Test for Homogeneity of Variance`
##   DFn DFd      SSn      SSd         F         p p<.05
## 1   2  74 45.12577 6928.918 0.2409689 0.7864803
```

The assumption is met.

Since normality and homogeneity are fine, let's run the two-way ANOVA!

We already actually ran it above, so we'll just pluck out the required output

```
res$ANOVA
```

```
##   Effect DFn DFd        F          p p<.05       ges
## 1  shelf   2  74 4.792833 0.01103308     * 0.1146807
```

$F(2, 74) = 4.79$, $p = .011$, so we reject H0 and conclude that there is a mean difference between groups.

We should also run post-hoc tests to see which groups differ

```
TukeyHSD(x = res$aov)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = formula(aov_formula), data = data)
##
## $shelf
##                                   diff        lwr       upr     p adj
## Middle Shelf-Bottom Shelf -11.1726122 -21.182271 -1.162954 0.0249984
## Top Shelf-Bottom Shelf     -0.9254068  -9.860102  8.009289 0.9667512
## Top Shelf-Middle Shelf     10.2472055   1.450329 19.044082 0.0183694
```

Tukey's HSD test suggests that there is a statistically significant difference between middle- and bottom-shelf cereals (bottom shelf has a 11.17 rating increase) and between top shelf and middle shelf cereals (top shelf has a 10.25 rating increase), but not between top and bottom shelf cereals (they only have a 0.93 difference).

Some useful output for APA summary:

```
cereal_final |>
  select(rating) |>
  describeBy(cereal_final$shelf)
```

```
##
##  Descriptive statistics by group
## group: Bottom Shelf
##        vars  n  mean    sd median trimmed   mad   min   max range skew kurtosis
## rating    1 20 46.15 13.56  43.93   44.84 11.42 28.74 74.47 45.73 0.68    -0.51
##          se
## rating 3.03
## ------------------------------------------------------------
## group: Middle Shelf
##        vars  n  mean    sd median trimmed  mad   min   max range skew kurtosis
## rating    1 21 34.97 13.95  31.23   33.69 11.9 18.04 64.53 46.49 0.79    -0.71
```

```
##          se
## rating 3.04
## ------------------------------------------------------------
## group: Top Shelf
##        vars  n  mean    sd median trimmed  mad   min  max range skew kurtosis
## rating    1 36 45.22 12.97   40.8   43.78 8.85 28.59 93.7 65.11 1.58     3.21
##          se
## rating 2.16
```

```
res[1]
```

```
## $ANOVA
##   Effect DFn DFd        F          p p<.05       ges
## 1  shelf   2  74 4.792833 0.01103308     * 0.1146807
```

```
res[2]
```

```
## $`Levene's Test for Homogeneity of Variance`
##   DFn DFd      SSn      SSd         F         p p<.05
## 1   2  74 45.12577 6928.918 0.2409689 0.7864803
```

APA summary (the ** indicate italics):

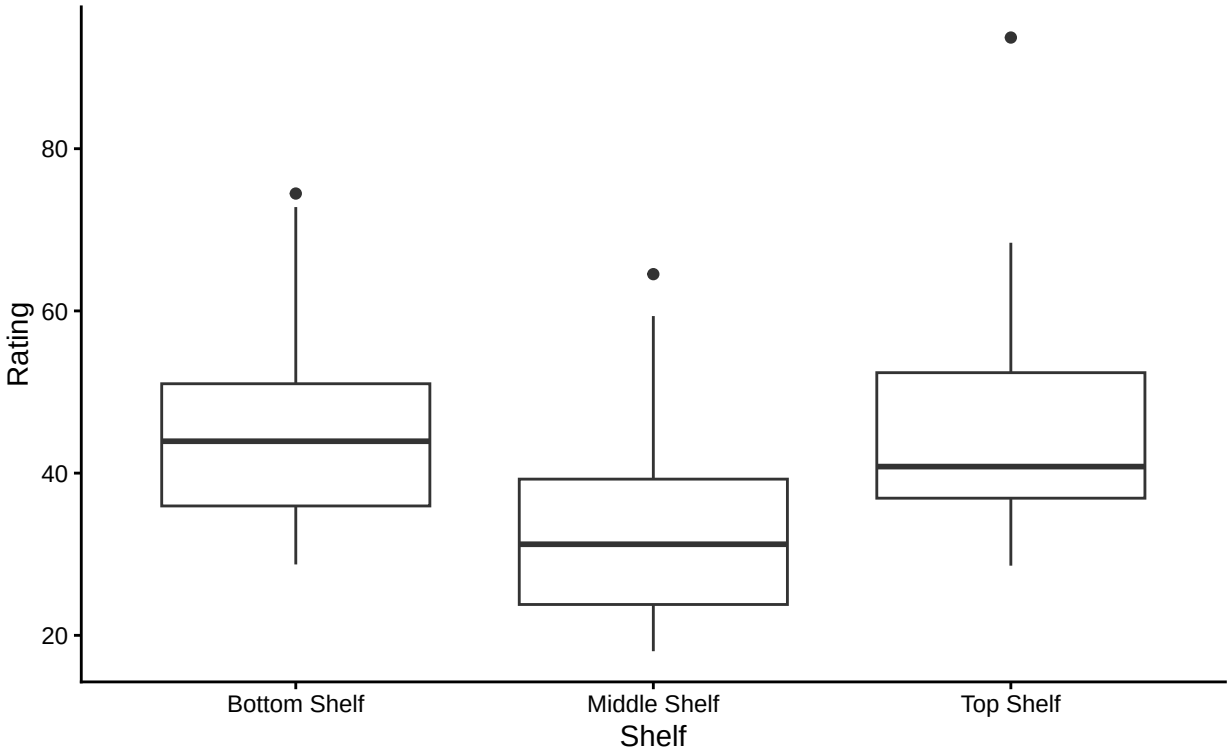
I wanted to determine if there was a statistically significant difference between popular cereal brands based upon where they tend to be placed on the shelf in stores (top, middle, or bottom-shelf).

The mean rating for bottom ($M = 46.15$, $SD = 13.56$) and top-shelf cereals ($M = 45.22$, $SD = 12.97$) was similar, but both were larger than middle-shelf cereals ($M = 34.97$, $SD = 13.95$). Figure 1 (below) depicts the distributional differences between the three shelf placements.

```
cereal_final |>
  ggplot(aes(x = shelf, y = rating)) +
  geom_boxplot() +
  theme_classic() +
  labs(x = "Shelf", y = "Rating",
       title = "Figure 1",
       subtitle = "Mean Rating Difference Between Shelf Placement of various Cereals")
```

Figure 1

Mean Rating Difference Between Shelf Placement of various Cereals



To test whether the means differed in the population, I ran a one-way ANOVA. Prior to doing so, however, I verified that several statistical assumptions were met, such as normality (as indicated by the roughly normal density curve, and close alignment of values to the theoretical Q-Q line) and homogeneity of variances (Levene's Test: $F(2, 74) = 0.24$, $p = .79$).

The ANOVA showed that there was a statistically significant rating difference between groups, $F(2, 74) = 4.79$, $p = .011$, $\eta^2 = 0.114$. Tukey's HSD tests confirmed that the mean differences between top- and middle-shelf, and bottom- and middle- shelves, were statistically significantly different.

The effect size of interest was raw mean differences, and the difference between middle- and other shelf-placements was roughly 10 rating units, which is considered quite large. Thus, it appears that middle- shelf cereals tend to have lower ratings compared to top- or bottom-shelf cereals.